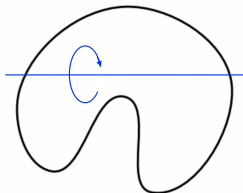


Classical Mechanics - Home Assignment 2

Calculus of Variations (August 2024)

1. A string of length L forms a closed loop on a plane. Find the shape of the loop for which the enclosed area is largest.
2. Consider a simple closed curve on a plane. We form a solid figure by revolving the curve about a straight line. For which shape of the curve the figure will have the largest volume for a fixed surface area?



3. Find the shortest path (geodesic) on the surface of a right circular cone.
4. Consider a curve C passing between two fixed end points (x_1, y_1) and (x_2, y_2) . We form a surface of revolution by revolving C about Y axis. Show that the area of this surface is minimum when C is a catenary.
5. A particle slides down a cycloidal track

$$x = a(\theta + \sin \theta), \quad y = a(1 - \cos \theta)$$

without friction under gravity. (Y axis is vertically upwards.) The particle starts from some arbitrary point (x_0, y_0) with zero initial velocity. Prove that the time required to slide to the point $(0, 0)$ is independent of the starting point. (Hence, a cycloid is called a *tautochrone* under constant force field.)

6. Consider two cycloids given by,

$$x = \pm a(\theta - \sin \theta), \quad y = a(1 - \cos \theta) \quad 0 \leq \theta \leq 2\pi$$

These cycloids made of wood are fixed next to each other on the ceiling of a room with all the cusps touching the ceiling. A simple pendulum with a flexible but inextensible string of length $L = 4a$ is suspended from the wedge-like meeting point of the two adjacent cusps (see figure). Show that the bob lies on the cycloids given by

$$x' = \pm a(\theta + \sin \theta), \quad y = a(3 + \cos \theta).$$

Hence prove that the pendulum swings with a period independent of the amplitude.

